1. GENERAL INFORMATION

Table 1. General information of the experimental plan.

General information	
Tittle of experimental plan:	EGB-250 reciprocating compressor fault detection
Number experimental plan:	EP_011V0
Director:	Eng. Vinicio Sánchez Loja, Ph.D.
Investigation:	Eng. Mariela Cerrada, Ph.D. / Eng. Diego Cabrera, Ph.D.
Mechanical assistant(s):	Franco Cajas / Cristian Torres / Holger Llivicura
Electrical assistant(s):	Eng. Jean Carlo Macancela / Eng. Mauricio Villacis
Supplier(s):	Eng. Fabián Pérez / Eng. Michael Wohlwend
Start date:	11/09/18
End date:	24/09/18
Locate:	Laboratory of Mechanical Engineering of the Universidad Politécnica Salesiana, Cuenca, Azuay, Ecuador.

2. DESCRIPTION

GIDTEC is an investigation group focused on fault detection and estimation analysis of mechanical systems by equipment condition signals, according to the parameters of Condition Based Maintenance. For this database it has a two-stage reciprocating compressor, an induction motor of 5.5HP, two pulleys with his V-Belts and an air tank of 250lt. In terms of measuring instruments will be present: a tachometer, four accelerometers, three current clamps, four pressure sensors and two condenser microphones, all are fed into a laptop through a data acquisition chassis (DAQ, NI cDAQ) in a program developed by members of the investigation group.

Generated signals with different faults modes in the critically elements of the reciprocating compressor are collected for a diagnosis of the system under study, where the signals taken with elements in healthy condition will be the starting point.

In the report some important features for the analysis are detailed, for example, the procedure for data acquisition and implant the faults in critically elements of compressor. The experimentation are high severity faults in critically elements of reciprocating compressor.

3. MECHANICAL ELEMENTS

Figure 1a shows the external parts and Figure 1b the main internal parts of the EBG250 compressor. Also, a diagram of mechanical operation is shown in Figure 2. In addition, Table 2 shows the main characteristics of the reciprocating compressor studied.

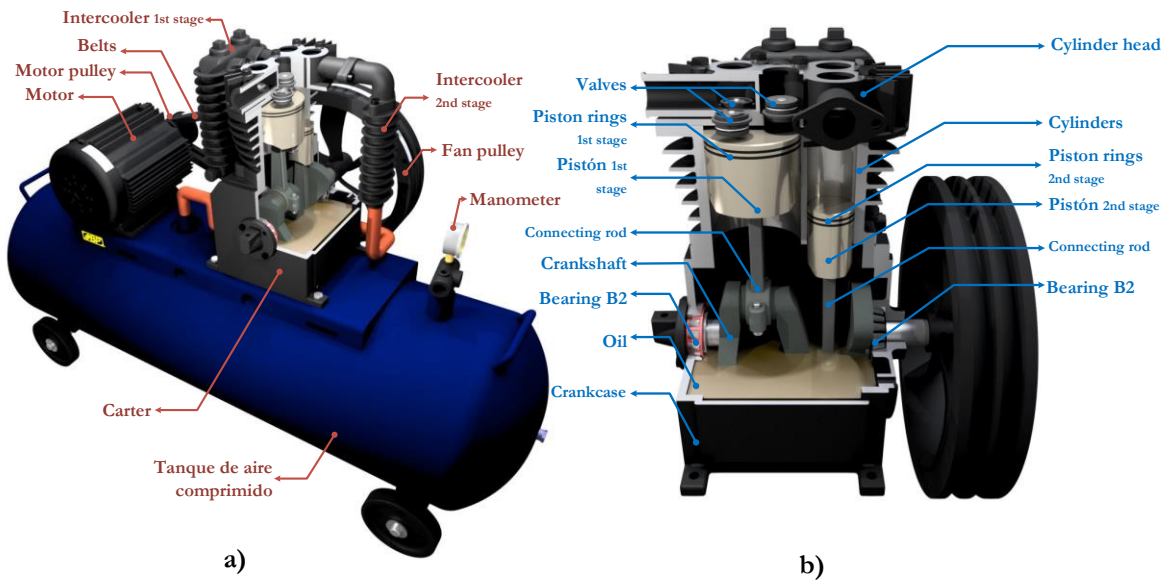


Figure 1. EBG250 Main parts of the EBG250 compressor: a) external parts b) Internal parts.

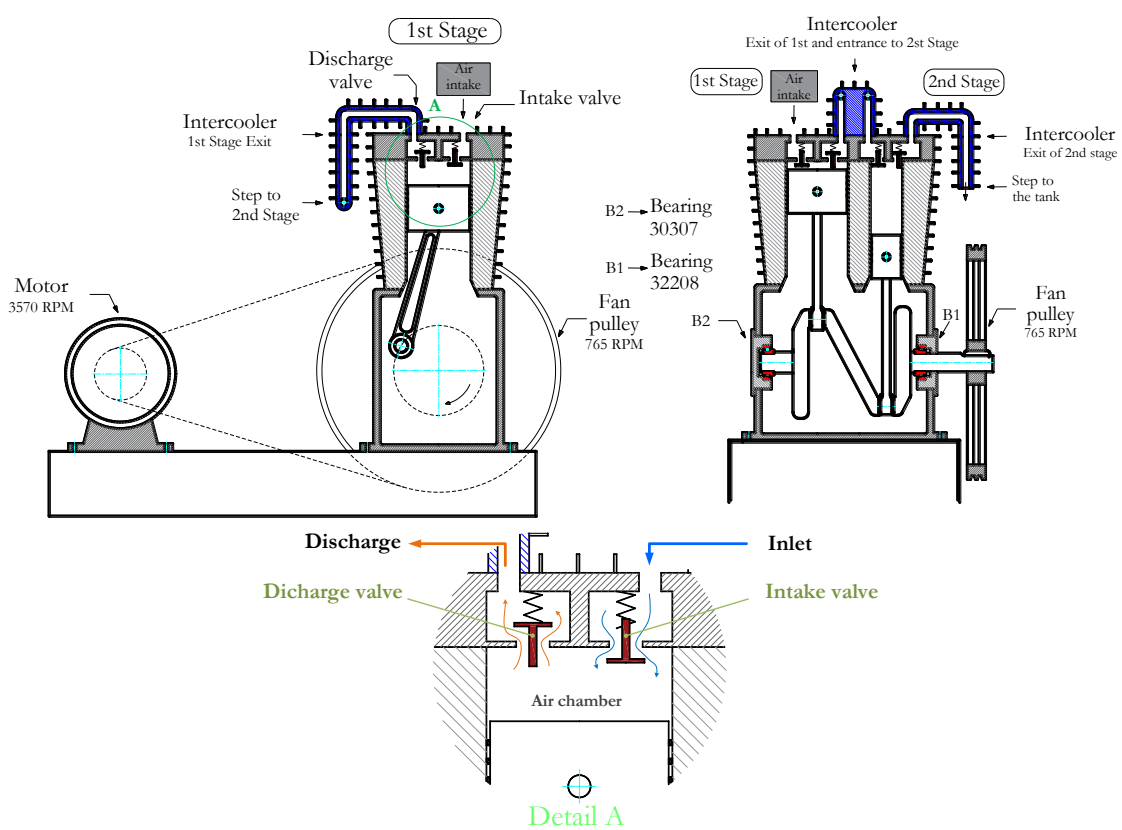


Figure 2. Scheme of the mechanical operation of the reciprocating compressor EBG250.

Table 2. Technical data compressor EBG250

Technical data of the EBG250 compressor	
Model:	EBG250
Type:	Reciprocating
Power:	5 HP (3.7 KW)
Motor:	TYPE YY112M-2
Motor power:	Three phase 360 V
Motor speed:	3470 rpm
Type of transmission:	Pulley -Belt
Reduction ratio:	4,5: 1
Type of belt:	V-belts
Ø motor pulley:	90 mm
Ø fan pulley:	420 mm
No. of stages:	2 stages (low and high pressure)
No. cylinders 1st stage:	1 (low pressure)
No. cylinders 2nd stage:	1 (high pressure)
Maximum pressure:	12 bars (175 psi)
Air flow:	24.5 CFM
Type of tank:	Horizontal tank
Tank capacity:	250 Liters
Filling time:	5 min
Total, compressor height:	1.24 m
Total, tank length:	1.79 m
Diameter tank:	85 cm
Type of lubrication:	Oil bath
Oil:	SAE 20W50 (GT-1 Kendall)
Oil quantity:	0,85 gal
Crankshaft rotation cycle time:	0,078 seconds

3.1 Bearings

The type of bearing used in the experiment is tapered roller bearings due to the ability to withstand high combined loads, are mounted on 'X' (face to face) one on each side of the crankshaft. The parts of the bearing are presented in Figure 3.

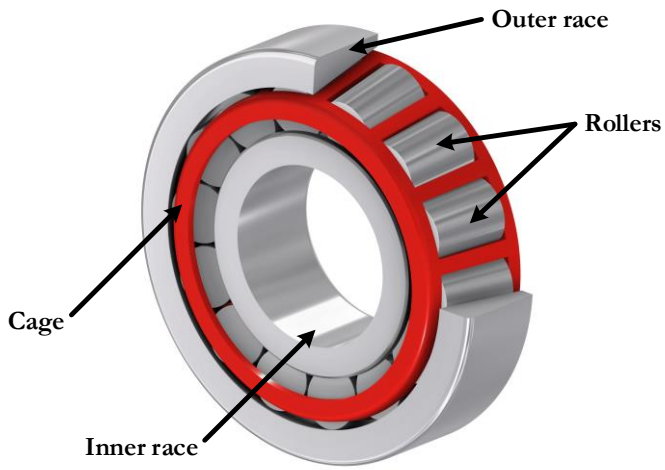


Figure 3. Parts of tapered roller bearing used in experimentation.

3.1.1 Features bearings

Main characteristics of bearings are presented in Table 3. In addition, in Figure 1 and Figure 2 it can be seen the location of each bearing inside the EGB250 reciprocating compressor.

Table 3. Features of compressor crankshaft bearings

Tapered Roller Bearings		
Internal code	**B1**	**B2**
ISO 355 denomination	32208	30307
Bore diameter	40mm	35mm
Outside diameter	80mm	80mm
Width	24,75mm	22,75mm
Number of rollers	17	14
Contact angle	13°25’	11°51’

3.2 Valves

The type of valve used for experimentation with the so-called single or individual plate type valve; its parts is presented in Figure 4a and the assembly form in Figure 4b.

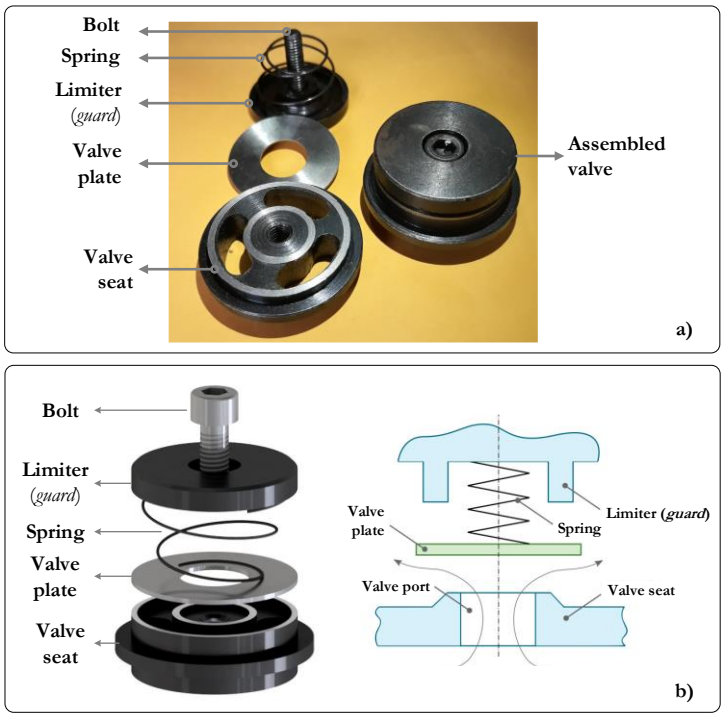


Figure 4. a) Parts of the valve; b) scheme of the assembly form.

Note: this valve works as an intake valve and as a discharge valve, its operation mode depends only on the orientation in which it is mounted. Figure 5 shows the form of orientation necessary to operate as admission or discharge.

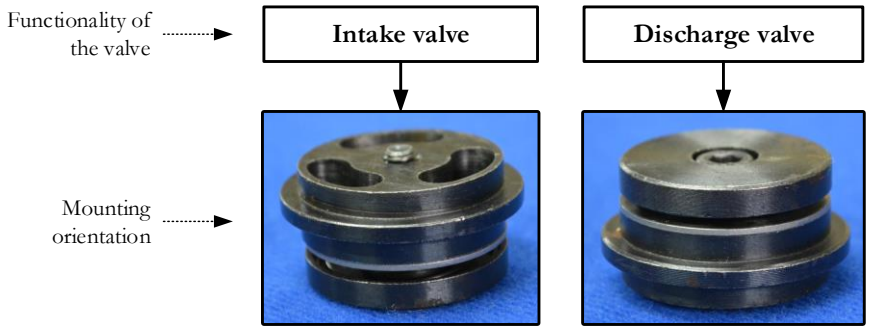


Figure 5. Valve mounting orientation.

3.2.1 Features valves

Main specifications of valve are presented in Table 4.

Table 4. Specifications of the valve used in the Reciprocating Compressor

Valve specifications		
Bolt	M6 x 40 mm	Material: Carbon steel
Limiter (guard)	External diameter = 34,5 mm Height = 5 mm	Material: Carbon steel
Helical Spring	Diameter wire= 0,7 mm Larger diameter = 32 mm Smaller diameter = 20 mm	Material: Carbon steel
Valve plate	External diameter = 34,5 mm Internal diameter = 14 mm Thickness= 1,05 mm	Material: Stainless steel
Valve seat	External diameter = 41,2 mm Height = 12,5 mm	Material: Carbon steel

3.2.2 Coding of the valves

Figure 6 shows the coding used for the identification of the EBG250 compressor valves. Finally, the location of the valves is detailed in Figure 7 and Figure 8.

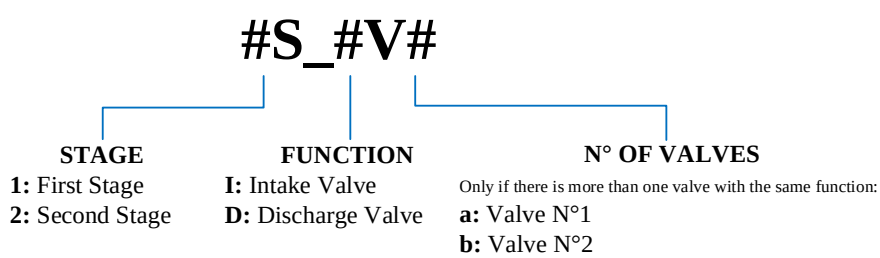


Figure 6. Valves codification.

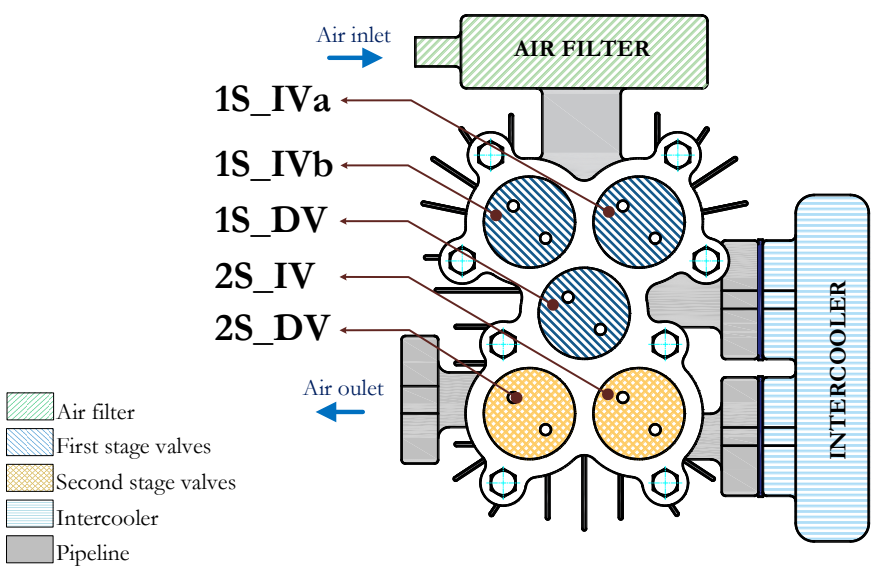


Figure 7. Identification of the position of the valves.

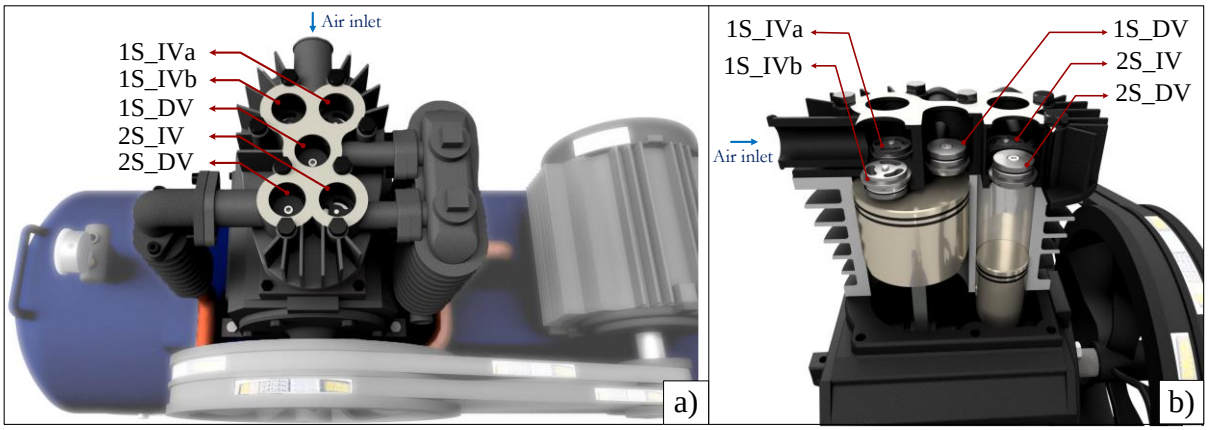


Figure 8. 3D Diagram of the location of the valves: a) Top view and b) view in section.

4. SETUP

4.1 Location of sensors

Figure 9 shows the schematic of the configuration of the sensors in the EBG250 compressor and in Figure 10 the location of the microphones is detailed.

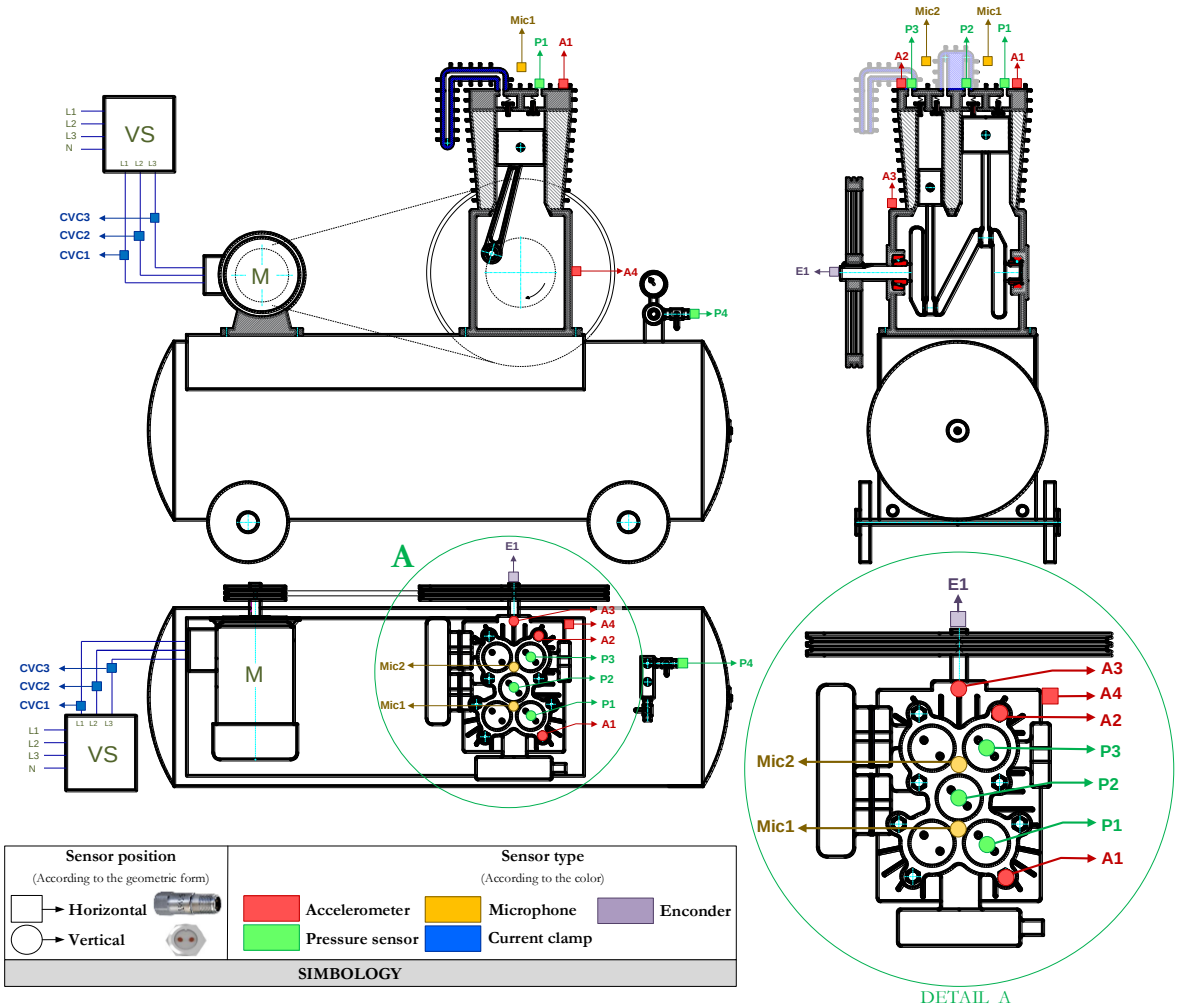


Figure 9. Schematic experimental set-up.

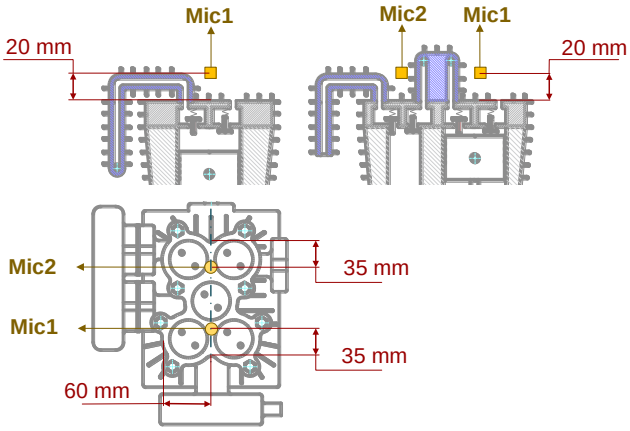


Figure 10. Location of microphones.

Additionally, Figure 11 shows the photographs of the sensor mounted on the EBG250 compressor.

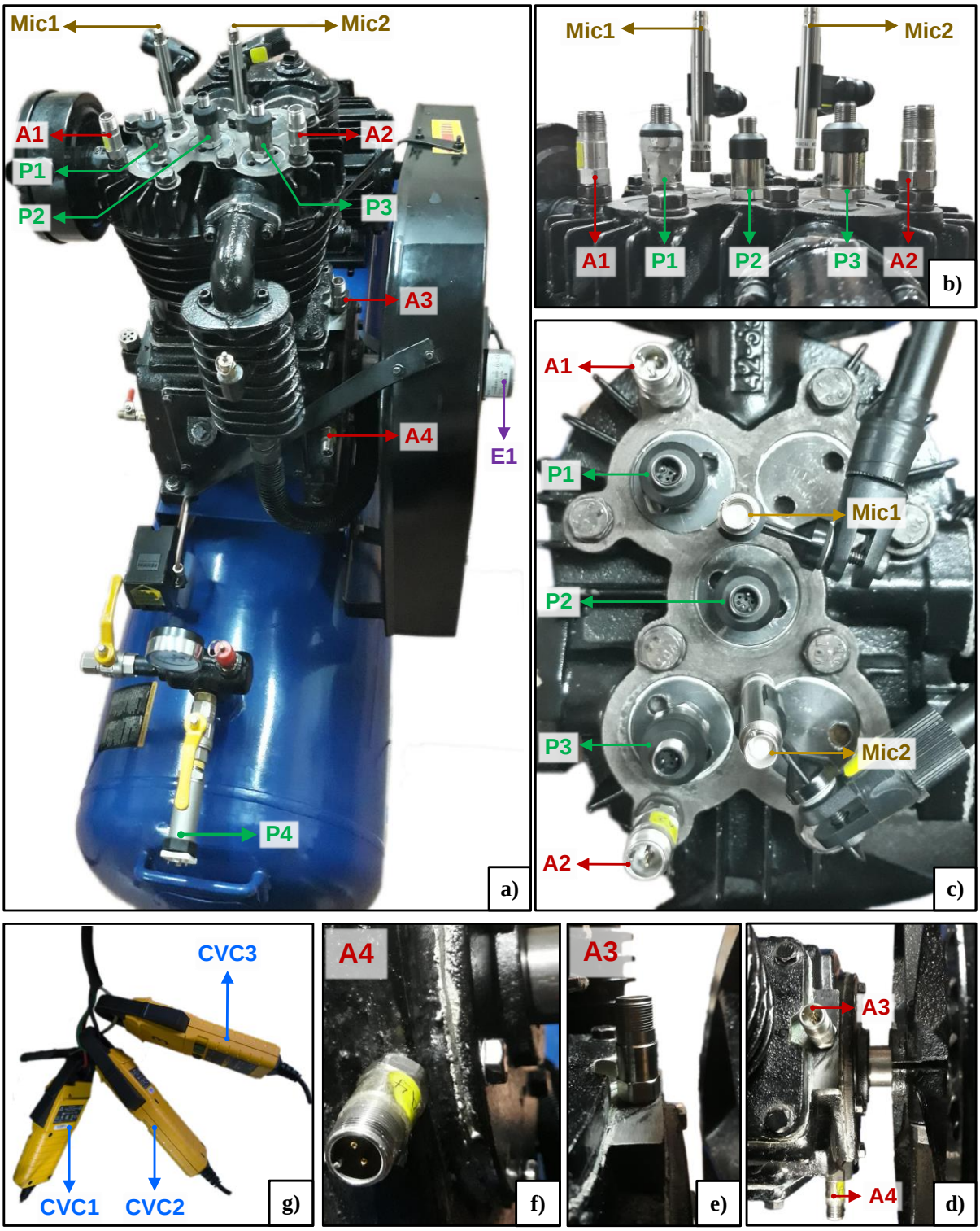


Figure 11. Sensors mounted on the compressor: a) complete view compressor; b) side view of the cylinder head; c) top view of the cylinder head; d) top view of the carter-crankshaft; e) placement of accelerometer A3; f) placement of accelerometer A4 and g) placement of ammeter clamps.

4.2 Configuration of sensors and modules

Figure 12 shows a diagram of the configuration of sensors and modules used in this experimental plan.

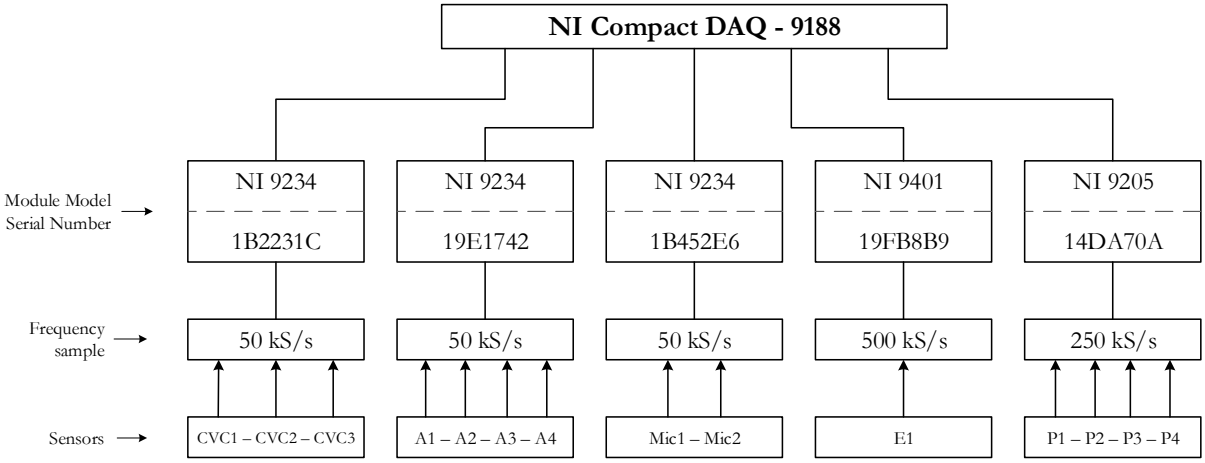


Figure 12. Setup of used sensors and modules

4.3 Channels directions

Table 5 details the characteristics of the channels of each sensor.

Table 5. Characteristics channels

Internal code	Sensor / Position	Channel Direction (Raw data)	Channel Direction	Brand	Model	Serial Number	Range	Sampling Frequency (kS/s)	Units of signal stored
P1	Pressure 1 / 1S_IVb*	data.Analog25k.TDMS Analog25kPressure0	data.Analog25k(1,:)	JUMO	MIDAS c08	213498501015420004	0 – 10 bar	25	Bar
P2	Pressure 2 / 1S_DV*	data.Analog25k.TDMS Analog25kPressure1	data.Analog25k(2,:)	JUMO	MIDAS c08	213498501015420002	0 – 10 bar	25	Bar
P3	Pressure 3 / 2S_DV*	data.Analog25k.TDMS Analog25kPressure2	data.Analog25k(3,:)	JUMO	MIDAS c08	213498501015420005	0 – 10 bar	25	Bar
P4	Pressure 4 / Air Tank	data.Analog25k.TDMS Analog25kPressure3	data.Analog25k(4,:)	JUMO	dTRANS p30	176027001012490001	0 – 10 bar	25	Bar
MC1	Microphone 1 / 1st Stage valves Vertical	data.Analog50k.TDMS Analog50kMic0	data.Analog50k(1,:)	PCB	HT378B02	135418	0 - 138,5dB	50	Pa
MC2	Microphone 2 / 2nd Stage valves Vertical	data.Analog50k.TDMS Analog50kMic1	data.Analog50k(2,:)	PCB	HT378B02	135419	0 - 138,5dB	50	Pa
CVC1	Current Clamp 1 / Power line 1	data.Analog50k.TDMS Analog50kCurrentClamp0	data.Analog50k(3,:)	FLUKE	i30s	34340055	5mA – 30A AC/DC	50	V
CVC2	Current Clamp 2 / Power line 2	data.Analog50k.TDMS Analog50kCurrentClamp1	data.Analog50k(4,:)	FLUKE	i30s	35460167	5mA – 30A AC/DC	50	V

Internal code	Sensor / Position	Channel Direction (Raw data)	Channel Direction	Brand	Model	Serial Number	Range	Sampling Frequency (kS/s)	Units of signal stored
CVC3	Current Clamp 3 / Power line 3	data.Analog50k.TDMS Analog50kCurrentClamp2	data.Analog50k(5,:)	FLUKE	i30s	35040027	5mA – 30A AC/DC	50	V
A1	Accelerometer 1 / 1S_IV Vertical*	data.Analog50k.TDMS Analog50kAcelerometer0	data.Analog50k(6,:)	IMI SENSORS	603C01	LW274628	±50g	50	m/s²
A2	Accelerometer 2 / 2S_DV Vertical*	data.Analog50k.TDMS Analog50kAcelerometer1	data.Analog50k(7,:)	IMI SENSORS	603C01	LW271408	±50g	50	m/s²
A3	Accelerometer 3 / B1 Radial Vertical*	data.Analog50k.TDMS Analog50kAcelerometer2	data.Analog50k(8,:)	IMI SENSORS	603C01	LW271290	±50g	50	m/s²
A4	Accelerometer 4 / B2 Radial Horizontal*	data.Analog50k.TDMS Analog50kAcelerometer3	data.Analog50k(9,:)	IMI SENSORS	603C01	LW274818	±50g	50	m/s²
E1	Tachometer / Crankshaft	data.Analog500k.TDMS Analog500kEncoder0	data.Analog500k(15,:)	Sink	DFS60B-S4PL	1036723 1120-0798	4.5 – 30V	500	---

*Note: The specification of internal codes of elements used in column 'Sensor / Position' for accelerometers and pressure sensors can be seen in section 3.

5. SIGNAL COLLECTION

1) Length of each sample: 10 seconds
2) Repetition of the tests: 15 times for each test (R1, R2, R3, … R15)
3) Motor rotation frequency: 57,7 Hz (3462 rpm)
4) Crankshaft rotation frequency: 12,8 Hz (768 rpm)
5) Tank pressure: 3 bar
6) Diagnosis_Manual_Compresor_DataBase11_V1.0

6. OPERATING CONDITIONS

To guarantee the repeatability and reproducibility of the data acquisition, the operating conditions of the EBG250 compressor must be defined and standardized. Thus, the mechanical parameters of operation under normal conditions should have been verified first. The values of these parameters have been presented in Table 2.

If the mechanical parameters are correct, then the environmental conditions of the laboratory where the experiments are performed should be verified. The verification must be carried out before starting the start-up of the compressor and during the acquisition of the data. The values of the environmental parameters that serve as a reference for the operation of the compressor under normal conditions are presented in Table 6. In case these values are not within the established range, the data acquisition is not valid, that is, the test should be repeated when the environmental conditions allow it.

Table 6. Values of the environmental parameters required for the operation of the EBG250 compressor.

Parameter	Unity	Range	Value	Observations
Room temperature	°C	Min.	14	These parameters must be measured at least 1 meter around the compressor.
		Max.	24	
RH	-	Min.	42%	
		Max.	60%	
Environmental noise before starting the compressor	dBA	Max.	48	
Environmental noise during experimentation	dBA	Max.	80	

In addition, when the compressor is operational, temperature and pressure values must also be monitored during data acquisition. In Table 7, the values of the parameters that must be monitored during the operation of the compressor under normal conditions are presented.

Table 7. Values of the physical parameters of the compressor during acquisition

Parameter	Units	1st stage	2nd stage	Tank
Pressure	bar	-	-	2,9 to 3,1
Inlet pressure	bar	- 0,1 to 0,1	Variable	-
Discharge pressure	bar	1,3 to 2,5	2,9 to 4,1	-
Surface temperature in the intake valve cover	°C	37 to 65	38 to 72	-
Surface temperature in the discharge valve cover	°C	42 to 50	40 to 62	-

These values should serve as a reference to control the compressor. If during the acquisition of data any of these values is not within the range, the following measures should be taken as first solutions:

- To control that the pressure in the tank is within the established range, the opening of the ball valve of the compressed air outlet must be manually adjusted, checking in the manometer placing next to this valve the pressure stabilizes at 3 bars.
- If the temperature values are still below their minimum limit, you will probably have to keep the compressor on for a longer period, until the values reach their established range.
- If the temperature values have exceeded their maximum limit, you should probably turn off the compressor and let it rest until the values are within the established range.

The equipment and instrumentation necessary to record the established parameters is the following: thermostat, pyrometer, sound level meter, pressure sensors installed in the valve covers and the pressure gauge installed in the compressed air tank.

7. DATABASE

7.1 Table of bearing faults for experiment

To ensure the correct assembly of elements in the compressor, each fault modes are encoded as shown in Figure 13. In the Annexes section is presented an enlarged detail of the construction of each element fault mode.

Table 8 show the faults codes configured in reciprocating compressor for the data acquisition according to a preliminary analysis for establish the experimentation.

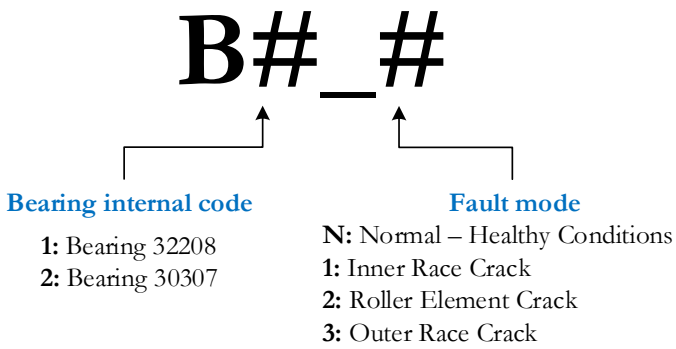


Figure 13. Fault modes codification for bearing.

Fault mode in tapered roller bearing:

- Normal – Healthy Conditions (N)
- Inner Race Crack (1)
- Roller Element Crack (2)
- Outer Race Crack (3)

Table 8. Codes for BEARING FAULTS.

BEARING FAULTS		
Fault code	**B1**	**B2**
P1	N	N
P2	B1_1	N
P3	B1_2	N
P4	B1_3	N

7.2 Table of valve faults for experiment

Each valve failure mode is coded as shown in Figure 14. In addition, Table 9 shows the configuration of the failures for the experimentation in valves. The detail of the construction of the failure mode of each element is presented in the annexes.

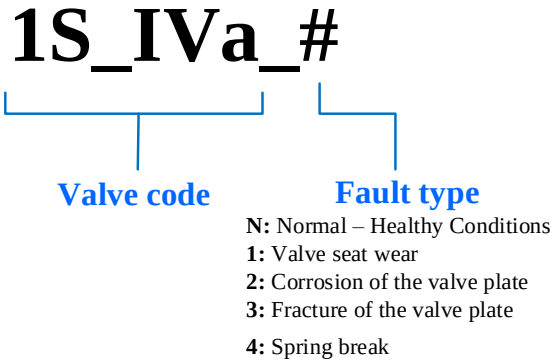


Figure 14. Valves fault modes codification.

Table 9. Codes for VALVES FAULTS.

VALVES FAULTS					
Fault code	**1S_IVa**	**1S_IVb**	**1S_DV**	**2S_IV**	**2S_DV**
P1	N	N	N	N	N
P2	N	N	N	N	2S_DV_1
P3	N	N	N	N	2S_DV_2
P4	N	N	N	N	2S_DV_3
P5	N	N	N	N	2S_DV_4
P6	N	N	N	2S_IV_1	N
P7	N	N	N	2S_IV_2	N
P8	N	N	N	2S_IV_3	N
P9	N	N	N	2S_IV_4	N
P10	N	N	1S_DV_1	N	N
P11	N	N	1S_DV_2	N	N
P12	N	N	1S_DV_3	N	N
P13	N	N	1S_DV_4	N	N
P14	N	1S_IVb_1	N	N	N
P15	N	1S_IVb_2	N	N	N
P16	N	1S_IVb_3	N	N	N
P17	N	1S_IVb_4	N	N	N
P18	N	N	N	2S_IV_1	2S_DV_1
P19	N	N	N	2S_IV_2	2S_DV_2
P20	N	N	N	2S_IV_3	2S_DV_3
P21	N	N	N	2S_IV_4	2S_DV_4
P22	N	1S_IVb_1	N	N	2S_DV_1
P23	N	1S_IVb_2	N	N	2S_DV_2
P24	N	1S_IVb_3	N	N	2S_DV_3
P25	N	1S_IVb_4	N	N	2S_DV_4

7.3 Table of multi- faults for experiment

Table 10 shows the fault codes configured in the compressor for the acquisition of data for multiple fault experimentation, which involves the combination of bearing and valves faults.

Table 10. Codes for MULTI-FAULTS.

MULTI- FAULTS							
	VALVES					BEARINGS	
Fault code	**1S_IVa**	**1S_IVb**	**1S_DV**	**2S_IV**	**2S_DV**	**B1**	**B2**
P1	N	N	N	N	N	N	N
P2	N	N	N	N	2S_DV_1	B1_1	N
P3	N	N	N	N	2S_DV_2	B1_1	N
P4	N	N	N	N	2S_DV_3	B1_1	N
P5	N	N	N	N	2S_DV_4	B1_1	N
P6	N	N	N	N	2S_DV_1	B1_2	N
P7	N	N	N	N	2S_DV_2	B1_2	N
P8	N	N	N	N	2S_DV_3	B1_2	N
P9	N	N	N	N	2S_DV_4	B1_2	N
P10	N	N	N	N	2S_DV_1	B1_3	N
P11	N	N	N	N	2S_DV_2	B1_3	N
P12	N	N	N	N	2S_DV_3	B1_3	N
P13	N	N	N	N	2S_DV_4	B1_3	N

7.4 Dataset

Table 11, Table 12 and Table 13 presents the list of parameters combinations for the data acquisition of the compressor, the number of fault codes depends to the faulty element.

RAW DATA

Table 11 . BEARING FAULTS database required for the experiment.

BEARINGS FAULTS		
Channels	**Fault codes**	**Repetition**
All *	P1	R1
	P2	R2
	P3	R3
	P4	R4
		R5
		R6
		R7
		R8
		R9
		R10
		R11
		R12
		R13
		R14
		R15

Table 12 . VALVES FAULTS database required for the experiment.

VALVES FAULTS		
Channels	**Fault codes**	**Repetition**
All *	P1	R1
	P2	R2
	P3	R3
	P4	R4
	P5	R5
	P6	R6
	P7	R7
	P8	R8
	P9	R9
	P10	R10
	P11	R11
	P12	R12
	P13	R13
	P14	R14
	P15	R15
	P16	
	P17	
	P18	
	P19	
	P20	
	P21	
	P22	
	P23	
	P24	
	P25	

Table 13 . MULTI-FAULTS database required for the experiment.

MULTI- FAULTS		
Channels	**Fault codes**	**Repetition**
All *	P1	R1
	P2	R2
	P3	R3
	P4	R4
	P5	R5
	P6	R6
	P7	R7
	P8	R8
	P9	R9
	P10	R10
	P11	R11
	P12	R12
	P13	R13
		R14
		R15

*Note: Experimentation acquires data on all channels (all sensors).

7.4.1 Dataset SUMMARY

Table 14, Table 15 and Table 16 present a summary of the tests.

Table 14. Summary data base BEARING FAULTS.

Number of folders	4 (P1 … P4)
Size each folder	540 MB
Files per folder	15
Size each file	36.8 MB
Number of files on this dataset	60
Size database	2.10 GB
Name of raw database:	Raw_Data_ Bearings_Faults_DB_011V0
Name of patched database:	Parche_Raw_Data_ Bearings_Faults_DB_011V0

Table 15. Summary data base VALVES FAULTS.

Number of folders	23 (P1 … P23)
Size each folder	538 MB
Files per folder	15
Size each file	35.9 MB
Number of files on this dataset	345
Size database	13.1 GB
Name of raw database:	Raw_Data_ Valves_Faults_DB_011V0
Name of patched database:	Parche_Raw_Data_ Valves_Faults_DB_011V0

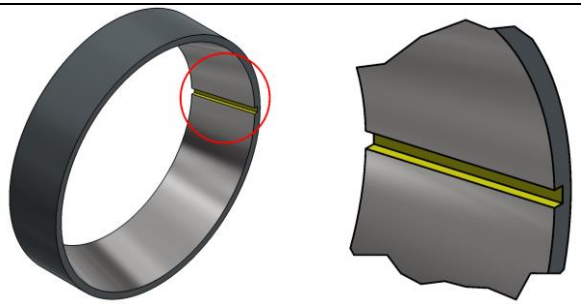
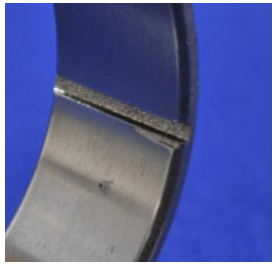
Table 16. Summary data base MULTI-FAULTS

Number of folders	13 (P1 … P13)
Size each folder	539 GB
Files per folder	15
Size each file	35.9 MB
Number of files on this dataset	195
Size database	6.86 GB
Name of raw database:	Raw_Data_ Multi_Faults_DB_011V0
Name of patched database:	Parche_Raw_Data_ Multi_Faults_DB_011V0

8. ANNEXES

8.1 Faulty patterns

Table 17. Fault patterns in BEARING.

Bearing Position	Coding faults in bearing	Fault mode	Fault description	Scheme or drawing	Photo
B1	N	Normal	Healthy condition		
B1	B1_1	Crack	EDM Technique Position: aligned to the rotation axis Length: All the raceway Width: 2,00 mm Depth: 1,00 mm		

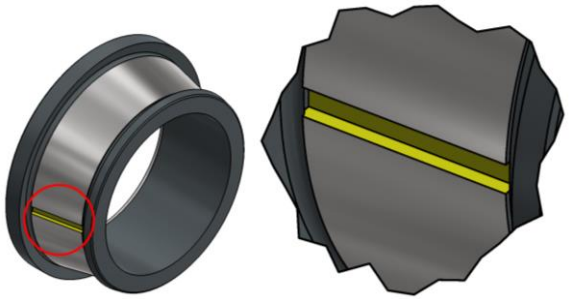
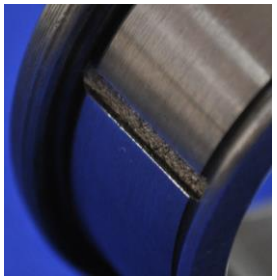
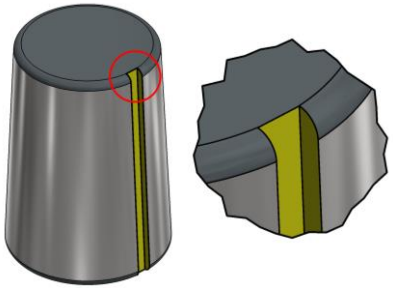
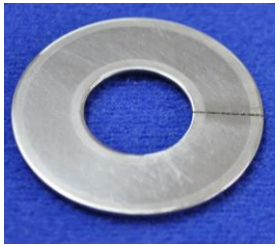
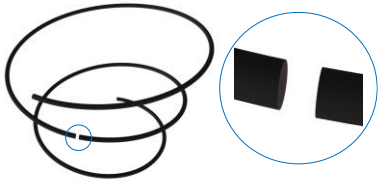
Bearing Position	Coding faults in bearing	Fault mode	Fault description	Scheme or drawing	Photo
B1	B1_2	Crack	EDM Technique Position: aligned to the rotation axis Length: All the raceway Width: 2,00 mm Depth: 1,00 mm		
B1	B1_3	Crack	EDM Technique Position: aligned to the rotation axis Length: All the raceway Width: 1,00 mm Depth: 0,50 mm		

Table 18. Fault patterns in VALVES

Coding of valves faults	Fault mode	Fault description	Internal code	Scheme or drawing	Photo
N	Normal	Healthy condition	N		
1S_DV_1 2S_IV_1 2S_DV_1 1S_IVb_1	Valve seat wear	Machining on the face of the valve seat. Depth of roughing = 1.44 mm	WVS_3		

Report Information	Elaborated		Revised	Approved
Member(s)	Franco Cajas	Cristian Torres	Eng. Vinicio Sánchez, Ph.D.	Eng. Vinicio Sánchez, Ph.D.
Date	17/08/2018		01/10/2018	20/11/2018

Coding of valves faults	Fault mode	Fault description	Internal code	Scheme or drawing	Photo
1S_DV_2 2S_IV_2 2S_DV_2 1S_IVb_2	Corrosion of the valve plate	Through hole Diameter = 2.50 mm	CVP_3		
1S_DV_3 2S_IV_3 2S_DV_3 1S_IVb_3	Fracture of the valve plate	Cut by thread Thread diameter = 1.6 mm Length = Short total Depth = all thickness.	VPF_3		
1S_DV_4 2S_IV_4 2S_DV_4 1S_IVb_4	Spring break	Cut by thread Thread diameter = 1.6 mm Depth = all thickness.	VSF_3